



Université Chouaib Doukkali
Ecole Nationale des Sciences Appliquées d'El Jadida
Département Télécommunications, Réseaux et Informatique
Module : **MLops**



TP

Filière : **SDIA**

Niveau : **2^{ème} Année**

Lab 4 :

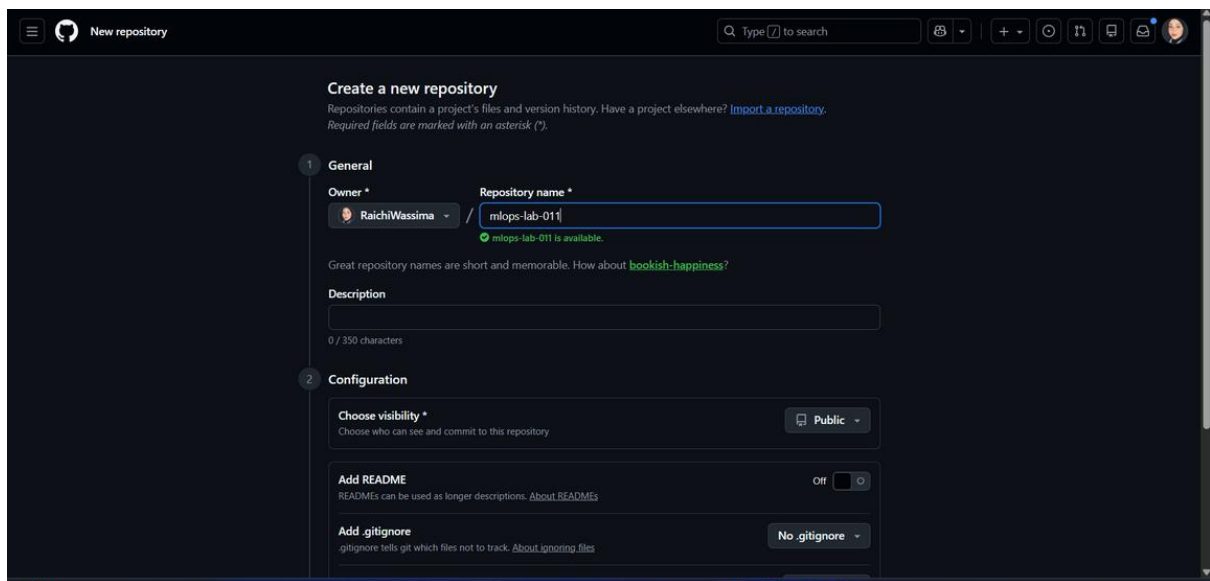
**Mise en place d'un pipeline CI/CD
complet pour un projet Machine
Learning**

Réalisé Par :
Wassima RAICHI

Année Universitaire : 2025/2026

Étape 1 : Créer le dépôt GitHub et connecter le remote

L'objectif de cette étape est de créer un dépôt GitHub centralisé afin d'héberger le code du projet et de servir de point d'entrée au pipeline CI/CD. Le projet local a été connecté au remote GitHub et synchronisé avec succès, permettant ainsi le versionnement, le partage du code et l'automatisation des futures étapes d'intégration et de déploiement.

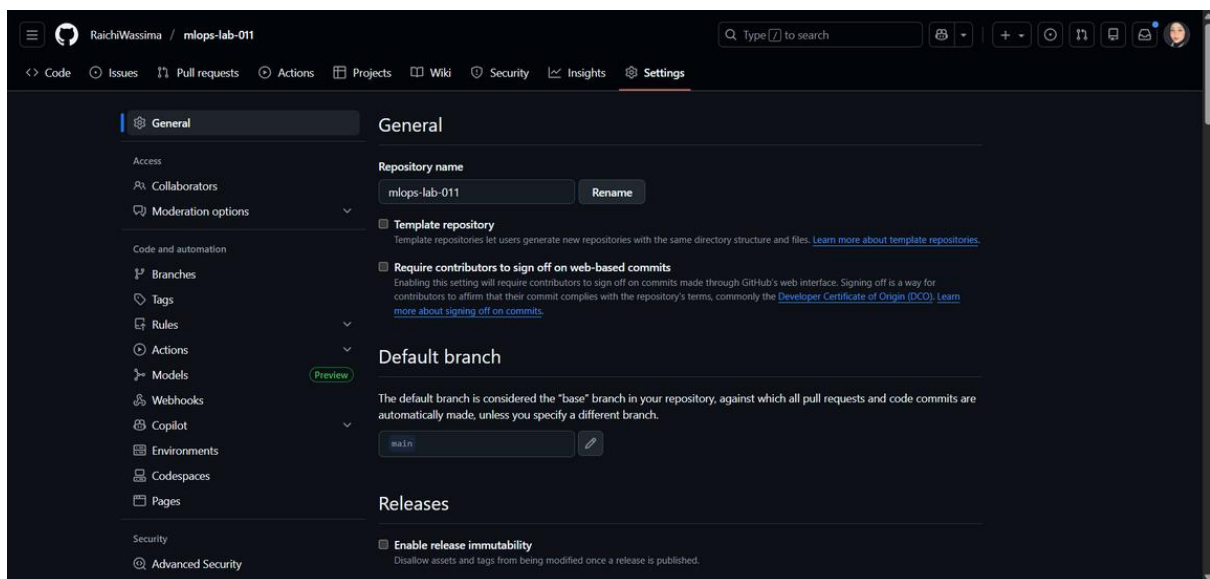
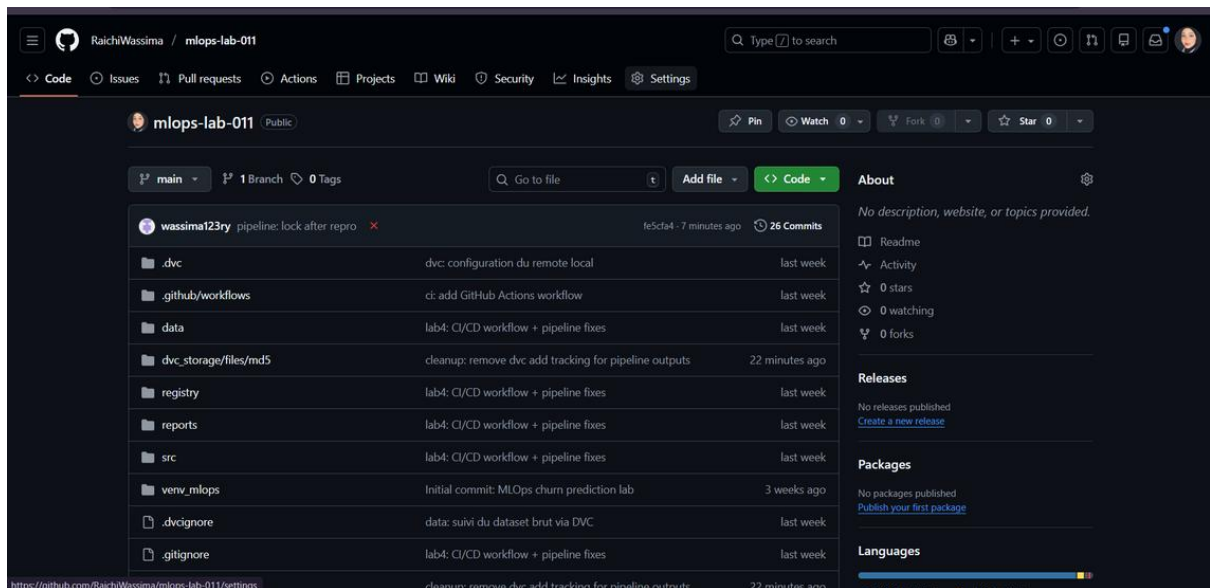


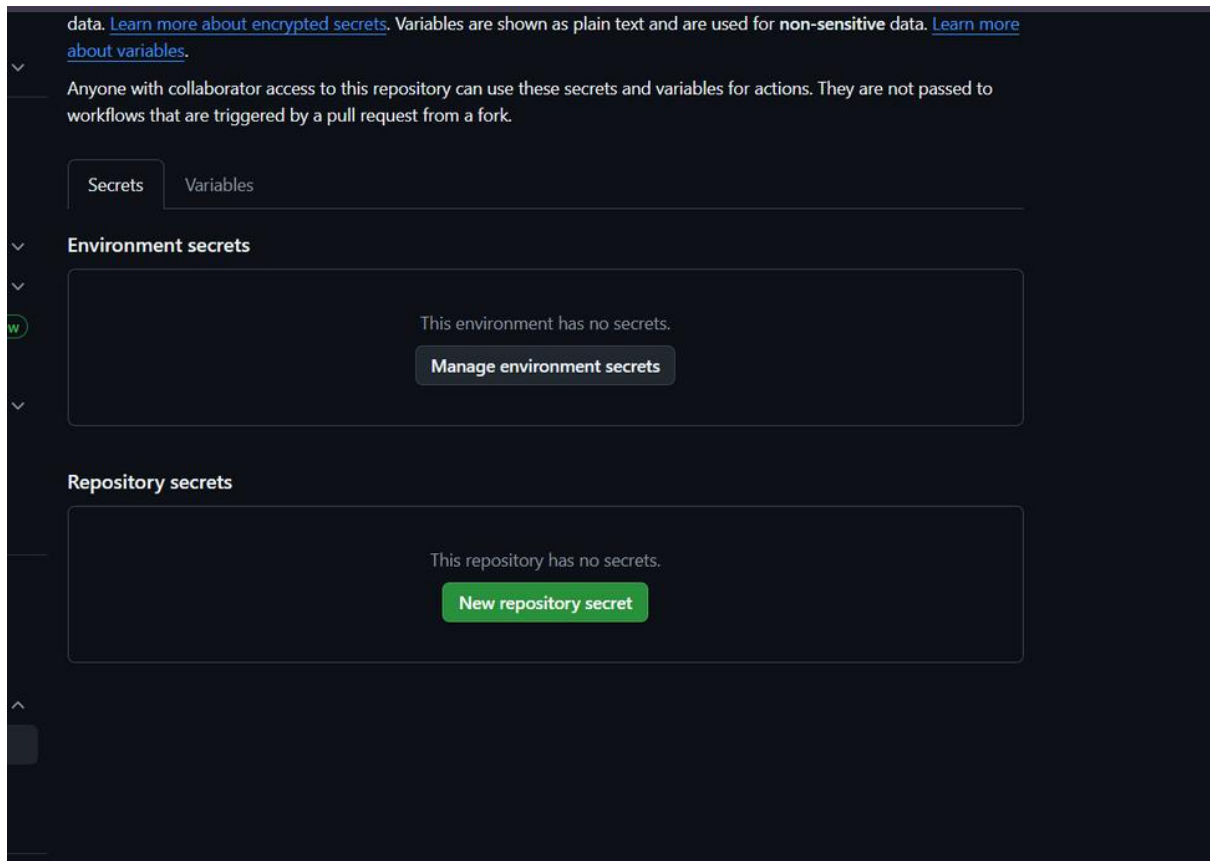
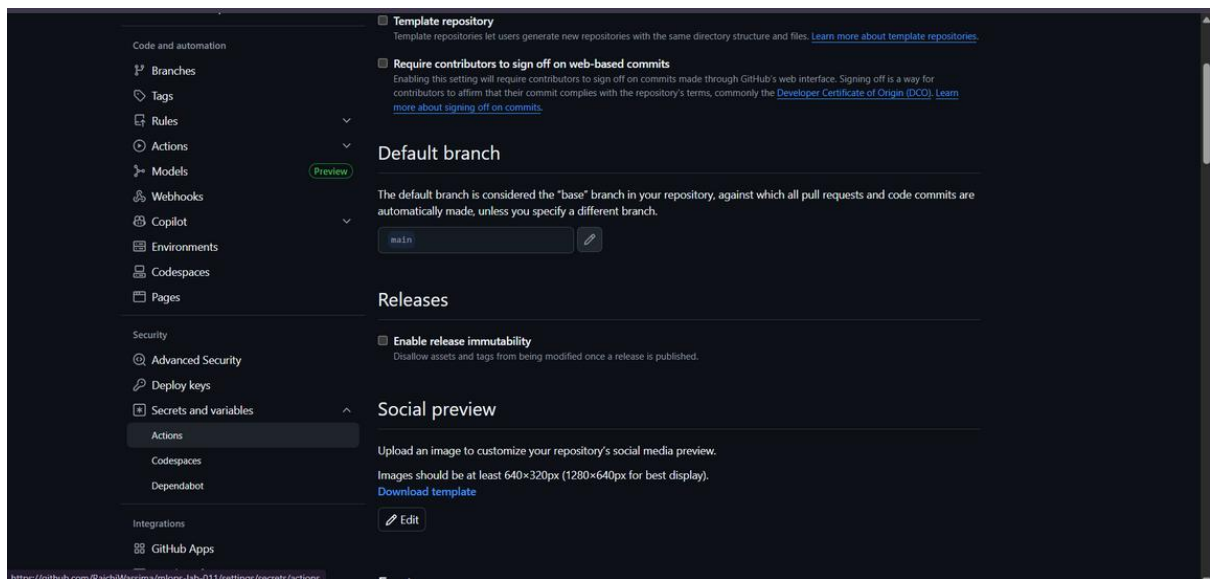
```
PS C:\Users\hp\OneDrive\Bureau\TP MLops\mlops-lab-011> git remote -v
origin https://github.com/RaichiWassima/mlops-lab-01.git (fetch)
origin https://github.com/RaichiWassima/mlops-lab-01.git (push)
PS C:\Users\hp\OneDrive\Bureau\TP MLops\mlops-lab-011> git remote set-url origin https://github.com/RaichiWassima/mlops-lab-011.git
PS C:\Users\hp\OneDrive\Bureau\TP MLops\mlops-lab-011> git push -u origin main
Enumerating objects: 7655, done.
Counting objects: 100% (7655/7655), done.
Delta compression using up to 8 threads
Compressing objects: 100% (6479/6479), done.
Writing objects: 100% (7655/7655), 70.68 MiB | 158.37 MiB/s, done.
Total 7655 (delta 1101), reused 7524 (delta 1052), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1101/1101), done.
To https://github.com/RaichiWassima/mlops-lab-011.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
PS C:\Users\hp\OneDrive\Bureau\TP MLops\mlops-lab-011> |
```

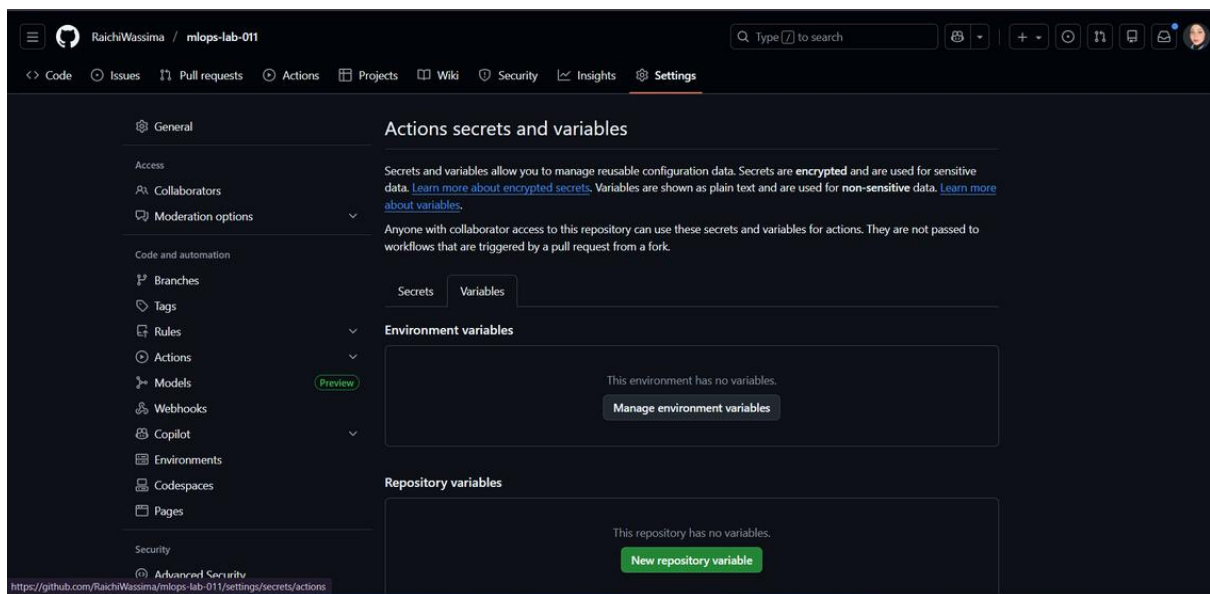
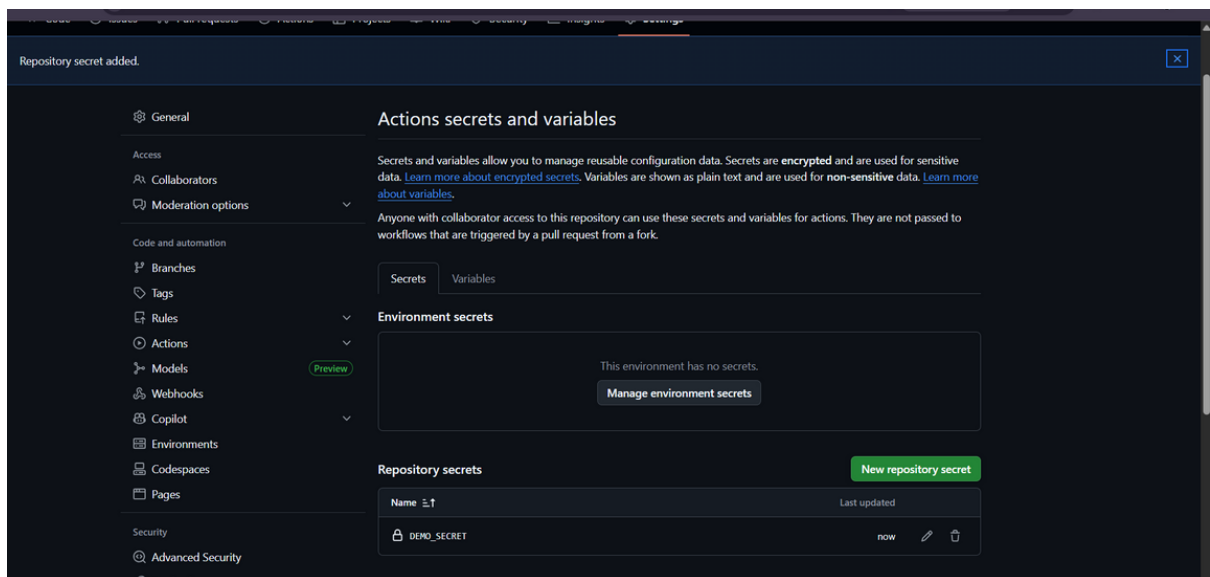
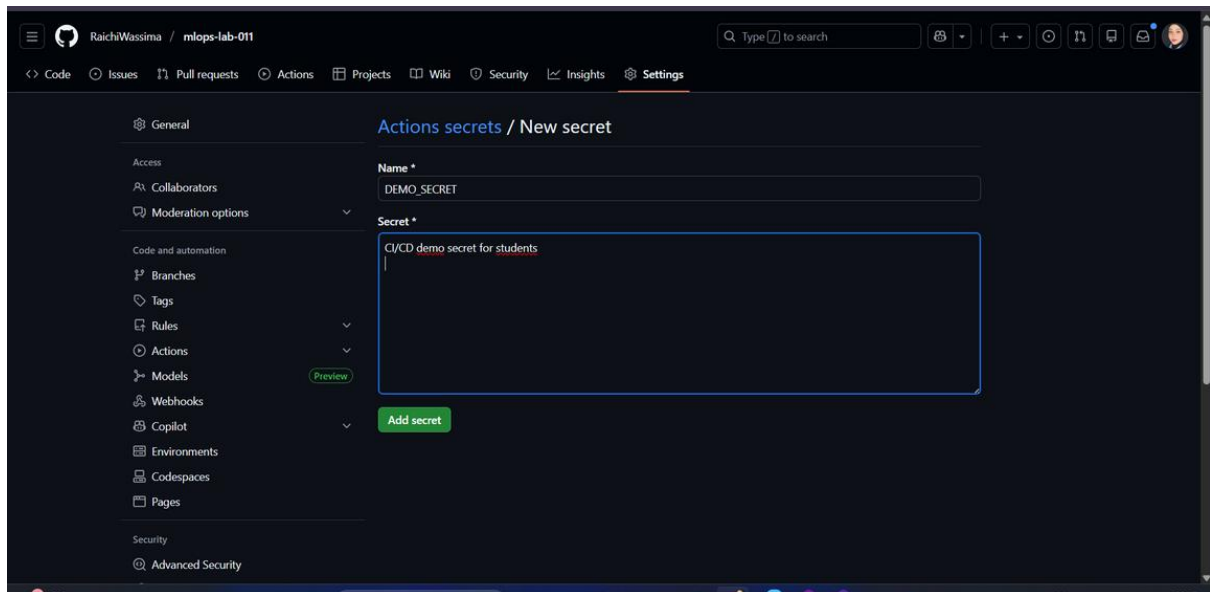
Étape 2 : Définir les secrets GitHub

L'objectif de cette étape est de sécuriser et paramétrer le pipeline CI/CD en externalisant les informations sensibles et les paramètres de configuration via les Secrets et Variables GitHub Actions. Les variables telles que la version de Python, le seuil de validation du modèle et l'environnement d'exécution, ainsi que les secrets confidentiels, sont désormais accessibles de manière sécurisée aux

workflows, garantissant un pipeline automatisé, configurable et conforme aux bonnes pratiques MLOps.







⋮

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Actions variables / New variable

Note: Variable values are exposed as plain text. If you need to encrypt and mask sensitive information, [create a secret](#) instead.

Name *

PY_VERSION

- Variable names may only contain alphanumeric characters ([a-z], [A-Z], [0-9]) or underscores (_).
- Variable names cannot start with a number.
- Variable names cannot start with GITHUB_ prefix.

Value *

3.10

Add variable

⋮

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Actions variables / New variable

Note: Variable values are exposed as plain text. If you need to encrypt and mask sensitive information, [create a secret](#) instead.

Name *

F1_GATE_THRESHOLD

- Variable names may only contain alphanumeric characters ([a-z], [A-Z], [0-9]) or underscores (_).
- Variable names cannot start with a number.
- Variable names cannot start with GITHUB_ prefix.

Value *

0.70

Add variable

⋮

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Actions variables / New variable

Note: Variable values are exposed as plain text. If you need to encrypt and mask sensitive information, [create a secret](#) instead.

Name *

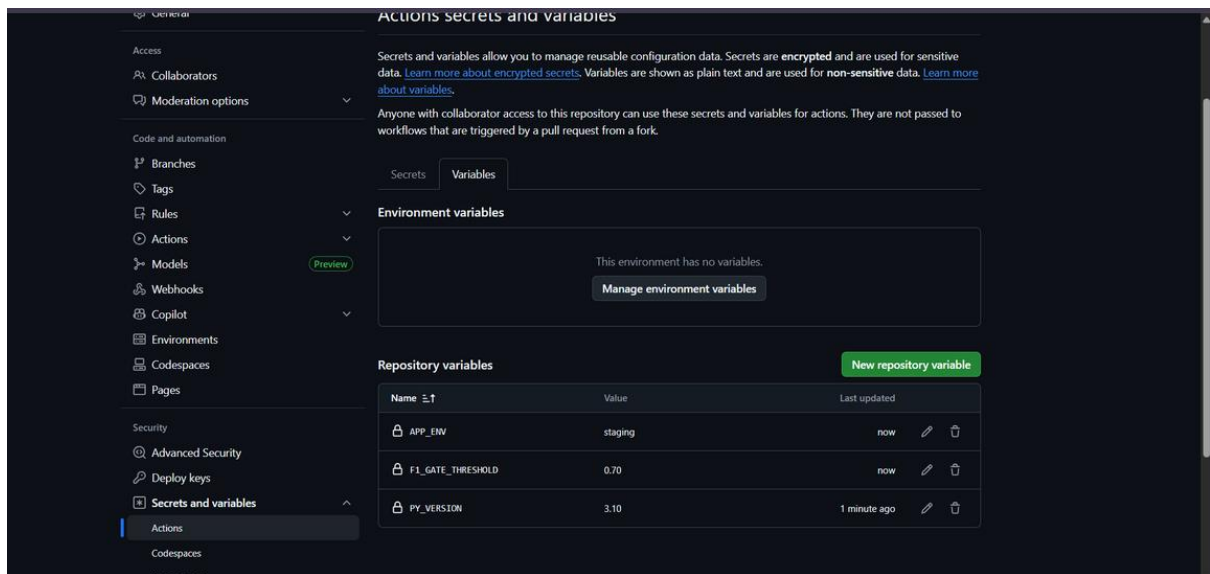
APP_ENV

- Variable names may only contain alphanumeric characters ([a-z], [A-Z], [0-9]) or underscores (_).
- Variable names cannot start with a number.
- Variable names cannot start with GITHUB_ prefix.

Value *

staging

Add variable

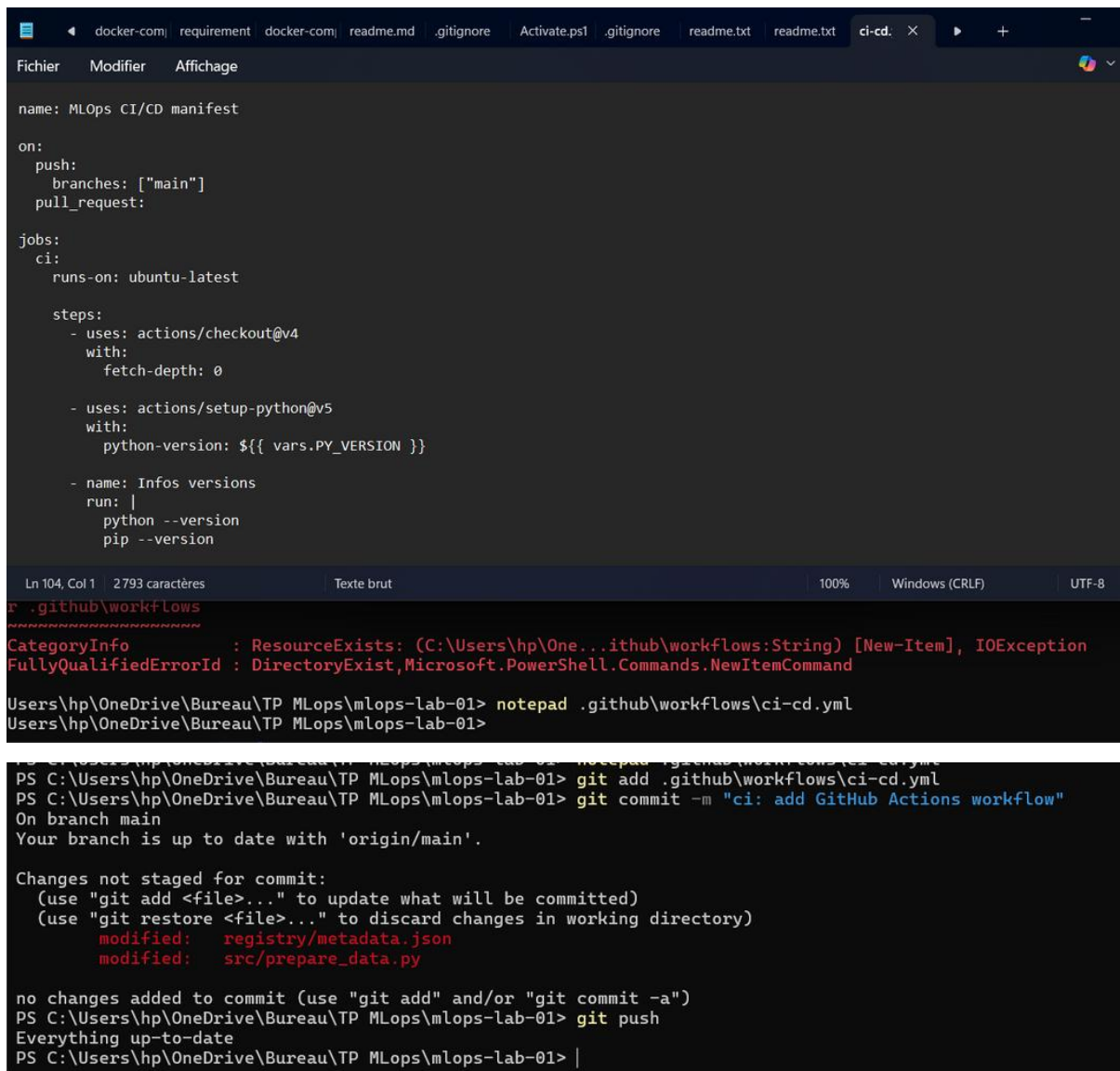


Étape 3 : Créer le workflow CI/CD

L'objectif de cette étape est d'automatiser l'intégration et le déploiement continu du projet de Machine Learning à l'aide de GitHub Actions, en définissant un workflow CI/CD déclenché à chaque push ou pull request. Le workflow mis en place permet d'installer l'environnement, d'exécuter le pipeline ML via DVC, de valider la qualité du modèle à l'aide d'un seuil de performance (Quality Gate) et de préparer les artefacts validés pour une phase de déploiement simulée, garantissant ainsi un pipeline MLOps fiable et automatisé.

```
PS C:\Users\hp\OneDrive\Bureau\TP MLops\mlops-lab-01> mkdir .github
mkdir : Il existe déjà un élément avec le nom spécifié C:\Users\hp\OneDrive\Bureau\TP MLops\mlops-lab-01\.github.
Au caractère Ligne:1 : 1
+ mkdir .github
+ ~~~~~
+ CategoryInfo          : ResourceExists: (C:\Users\hp\One...-lab-01\.github:String) [New-Item], IOException
+ FullyQualifiedErrorId : DirectoryExist,Microsoft.PowerShell.Commands.NewItemCommand

PS C:\Users\hp\OneDrive\Bureau\TP MLops\mlops-lab-01> mkdir .github\workflows
mkdir : Il existe déjà un élément avec le nom spécifié C:\Users\hp\OneDrive\Bureau\TP MLops\mlops-lab-01\.github\workflows.
Au caractère Ligne:1 : 1
+ mkdir .github\workflows
+ ~~~~~
+ CategoryInfo          : ResourceExists: (C:\Users\hp\One...ithub\workflows:String) [New-Item], IOException
+ FullyQualifiedErrorId : DirectoryExist,Microsoft.PowerShell.Commands.NewItemCommand
```

The screenshot shows a code editor with a dark theme. The top bar displays several open files: `docker-com| requirement docker-com| readme.md | .gitignore Activate.ps1 .gitignore readme.txt readme.txt ci-cd. x`. The active file is `ci-cd. x`, which contains a GitHub Actions workflow definition:

```
name: MLOps CI/CD manifest

on:
  push:
    branches: ["main"]
  pull_request:

jobs:
  ci:
    runs-on: ubuntu-latest

    steps:
      - uses: actions/checkout@v4
        with:
          fetch-depth: 0

      - uses: actions/setup-python@v5
        with:
          python-version: ${vars.PY_VERSION}

      - name: Infos versions
        run: |
          python --version
          pip --version
```

Below the code editor, a terminal window shows the following commands and output:

```
Ln 104, Col 1 | 2793 caractères | Texte brut | 100% | Windows (CRLF) | UTF-8
r .github\workflows
CategoryInfo          : ResourceExists: (C:\Users\hp\OneDrive\Bureau\TP MLops\mlops-lab-01\workflows:String) [New-Item], IOException
FullyQualifiedErrorId : DirectoryExist,Microsoft.PowerShell.Commands.NewItemCommand

Users\hp\OneDrive\Bureau\TP MLops\mlops-lab-01> notepad .github\workflows\ci-cd.yml
Users\hp\OneDrive\Bureau\TP MLops\mlops-lab-01>

PS C:\Users\hp\OneDrive\Bureau\TP MLops\mlops-lab-01> notepad .github\workflows\ci-cd.yml
PS C:\Users\hp\OneDrive\Bureau\TP MLops\mlops-lab-01> git add .github\workflows\ci-cd.yml
PS C:\Users\hp\OneDrive\Bureau\TP MLops\mlops-lab-01> git commit -m "ci: add GitHub Actions workflow"
On branch main
Your branch is up to date with 'origin/main'.

Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   registry/metadata.json
        modified:   src/prepare_data.py

no changes added to commit (use "git add" and/or "git commit -a")
PS C:\Users\hp\OneDrive\Bureau\TP MLops\mlops-lab-01> git push
Everything up-to-date
PS C:\Users\hp\OneDrive\Bureau\TP MLops\mlops-lab-01> |
```

Étape 4 : Commit et push

L'objectif de cette étape est de déclencher automatiquement le pipeline CI/CD en poussant les dernières modifications vers le dépôt GitHub. Le commit et le push ont permis de lancer le workflow GitHub Actions, au cours duquel le job CI a validé l'environnement, exécuté le pipeline ML et généré les artefacts, tandis que le job CD, exécuté uniquement sur la branche main, a simulé avec succès une phase de déploiement, confirmant le bon fonctionnement du pipeline CI/CD.


```
PS C:\Users\hp\OneDrive\Bureau\TP MLops\mlops-lab-01> git status
On branch main
Your branch is up to date with 'origin/main'.

Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   registry/metadata.json
        modified:   src/prepare_data.py

no changes added to commit (use "git add" and/or "git commit -a")
PS C:\Users\hp\OneDrive\Bureau\TP MLops\mlops-lab-01> |
```

Actions New workflow

All workflows

MLops CI/CD manifest

Management

Caches

Attestations

Runners

Usage metrics

Performance metrics

All workflows

Showing runs from all workflows

Filter workflow runs

1 workflow run

Event Status Branch Actor

ci: add GitHub Actions workflow

MLops CI/CD manifest #1: Commit a365022 pushed by RaichiWassima

main

1 minute ago

In progress

RaichiWassima / mlops-lab-01

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Actions variables / Update variable

Note: Variable values are exposed as plain text. If you need to encrypt and mask sensitive information, [create a secret](#) instead.

Name *

F1_GATE_THRESHOLD

Value *

0.65

Update variable

☰

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Workflow run deleted successfully.

Actions

New workflow

All workflows

MLOps CI/CD manifest

Management

Caches

Attestations

Runners

Usage metrics

Performance metrics

All workflows

Showing runs from all workflows

2 workflow runs

Event Status Branch Actor

ci: rerun workflow with updated F1 gate

MLOps CI/CD manifest #3: Commit [dac5fcd](#) pushed by RaichiWassima

main

🕒 14 minutes ago

🕒 1m 7s

...

lab4: CI/CD workflow + pipeline fixes

MLOps CI/CD manifest #2: Commit [cc1627f](#) pushed by RaichiWassima

main

🕒 24 minutes ago

🕒 1m 0s

...